



Mega-Terminal Programs That Underperformed: Year 1–3 Warning Signs

An Analytical Report Generated by the AI Research Council

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Abstract

Mega-terminal failures are not black swans; they are slow-motion governance failures with leading indicators visible 5–7 years before ribbon-cutting. The same three or four signals — scope creep, integrated-systems testing deferral, owner-engineer role confusion, and optimism bias in the business case — recur across BER, T5, LGA, and JFK T1. The report examines this claim across the following: I read it and go, "I never thought about it that way"; It has clear recommendations and examples.

How This Report Was Produced

This report is the product of the Transform Airports AI Research Council — a multi-agent analytical system running on Anthropic's Claude models. Rather than a single model writing in one pass, the Council convenes a panel of specialized research agents, subjects their work to adversarial review, and verifies every claim before publication. The process runs in four stages, with human review at two checkpoints.

Stage 1 — Independent research. Each research agent below produced an evidence brief without seeing the others' work, so the analysis draws on genuinely independent lines of evidence rather than a single line of reasoning.

Stage 2 — Synthesis and adversarial review. A Strategist synthesized the briefs into an argument; a Red Team attacked it across two rounds; the Strategist revised after each.

Stage 3 — Edit and fact-check. An Editor tightened the prose. A Fact-checker verified every number against the research briefs, removing or flagging what it could not source.

Stage 4 — Production. The verified draft was formatted into this document.

Research agents seated for this report

- Airline Commercial Strategist
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Process agents (on every report)

- **Strategist** — synthesized the research briefs into a single argumentative draft.
- **Red Team** — attacked each draft to surface weak claims and logical gaps.
- **Editor** — tightened the prose and removed filler without adding new content.
- **Fact-checker** — verified every numerical and attributed claim against the briefs.

Each agent is governed by a versioned specification — its analytical mandate, evidentiary standards, source discipline, and argumentative posture. The complete technical definitions for every agent on the Council are published and openly available at:

<https://github.com/TransformAirports/ai-council/tree/main/.claude/agents>

The Architecture of Silence

Mega-Terminal Programs That Underperformed, and the Warning Signs That Were Already in the Room

In May 1992, the engineers responsible for Munich Airport's automated baggage handling system put their warning on the record. They had just commissioned a system smaller and simpler than the one Denver was about to build, but new enough that the team had learned things worth saying out loud. The Munich system had taken two years to construct. After that came six months of round-the-clock testing — twenty-four hours a day, every day, for six months. Then it opened.

Denver had set itself a different schedule. The Denver system — seventeen miles of track, four thousand radio-controlled carts, one hundred networked computers, five thousand electric eyes — was an order of magnitude more complex than Munich's. The city's plan was to design, build, install, integrate, and test it in roughly the window Munich had spent on testing alone. Three companies had already told Denver the timeline was impossible. The city rejected all three bids and asked a fourth, BAE Automated Systems, to take the contract at the original price and schedule. BAE took it. Then the city expanded the scope from one concourse for one airline to three concourses for twenty.

Munich's warning reached Denver's project team. So did the warnings of the three contractors. So did the assessment of an outside consultant the city itself had hired, who called the plan "too complex" and the timeline impossible. Five independent, technically qualified warnings — Munich, three bidders, the consultant — with the names of engineers attached, arrived during the period when corrective action was still possible.

Denver International Airport opened in February 1995, sixteen months late. The Government Accountability Office documented approximately \$360 million in cumulative delay costs: \$230 million in operating deficits, \$37 million in lost income, \$86 million for the alternative baggage system, \$8 million in additional bond issuance. Monthly operating deficits ran \$13.35 million to \$18.75 million across the delay. The automated baggage system never worked at full design — it ran one concourse, one airline, outbound bags only — and was decommissioned in 2005 at a maintenance cost of a million dollars a month. United Airlines replaced it with a conventional manual system that could have been installed at the start for \$51 million.

What Munich's engineers wrote down in 1992 was a warning. It arrived on time. It was specific. It was technical. It was delivered by people with no incentive to be wrong. And it did not change the outcome.

This is the pattern that recurs across every major terminal failure of the last fifty years, and it is not what most post-mortems describe. The dominant story is that mega-terminal failures are black swans — rare, complex, unforeseeable. The honest story is the opposite. They are slow-motion governance failures with leading indicators visible five to

seven years before ribbon-cutting. The signals are not hidden. They are noticed, written down, sometimes formally delivered to people who could act on them. What fails is not detection. What fails is the institutional capacity to receive the signal and stop.

The warning signs are real, recurring, and identifiable — and also nearly universal, appearing in the great majority of megaprojects including the ones that opened well. Visibility is the wrong frame. The question a capital planner facing a multi-billion-dollar terminal program in 2026 needs to answer is not whether the signals exist. It is who in the organization has both the information and the standing to halt the program when they appear. The current US terminal wave is the largest concentration of mega-programs since 1989–1998. If the institutional architecture is not built before construction begins, the warnings will arrive and they will be filtered, softened, or politically inadmissible by the time they reach anyone who can act.

Executive Preface

The four leading indicators every post-mortem itemizes — scope creep without re-baselining, integrated systems testing compressed to protect a fixed opening date, owner-engineer role confusion, optimism bias in the business case — appear in nearly every documented mega-terminal failure. They also appear in nearly every megaproject that opened well. Bent Flyvbjerg's database of over 16,000 projects finds that only 8.5% meet time and budget targets. The signals are universal at the design phase. They are not, by themselves, predictors. What predicts the outcome is whether the institution receiving the signal has the standing to act on it — whether someone with operational accountability and board-level authority can refuse a political opening date that the systems work cannot meet.

The cost of getting that wrong is large and routine. BER's nine-year delay drove the program from roughly €2.8 billion to over €7 billion in direct spend, and to €10 billion or more when interest and lost revenue are included. The Denver Great Hall P3, contracted at \$650 million in 2018, was terminated in 2019 and is now projected at \$2.35 billion. The LAX Automated People Mover overran by 46% and absorbed \$750 million in dispute settlements. Fitch's published debt-service coverage bands place A-rated airports at 1.25x–1.50x; a \$300 million overrun on a \$1.5 billion terminal can push coverage below covenant and widen spreads by 30–60 basis points on the next issuance — roughly \$4.5–9 million annually on a \$1.5 billion deal, with a 30-year tail. Heathrow Terminal 5 recovered within eleven months and went on to win five consecutive Skytrax awards; BER still operates below the traffic level its retail revenue model assumed at opening.

The current US wave puts seven such programs into execution at once — JFK New Terminal One at \$9.5 billion, JFK Terminal 6 at \$4.2 billion, O'Hare expansion grown from \$8.5 billion to at least \$11.5 billion, the LAX automated people mover grown from \$1.9 billion to \$2.78–3.34 billion, the completed \$4 billion LaGuardia Terminal B, Delta's \$4 billion LaGuardia Terminal C, and DFW Terminal F at \$4 billion. NPIAS-

identified needs rose from \$43.5 billion in FY2021–2025 to \$62.4 billion in FY2023–2027, driven by 10.1% construction inflation in 2021, 14.8% in 2022, and expanded scope. The Infrastructure Investment and Jobs Act's \$19.5 billion supplemental aviation pool expires in December 2026; the baseline Airport Improvement Program then reverts to roughly \$4 billion annually. Most of these programs were business-cased under a different assumption set. The PFC has been frozen at \$4.50 since 2000 — worth \$2.72 in 2018 dollars. That single political fact is upstream of the financing structures that produce the governance fragmentation that produces the failures. The chain runs federal politics → financing structure → governance fragmentation → opening-day failure. Most post-mortems begin three links downstream.

The Argument

The Munich Warning

Sit with the institutional question. Denver received five technically qualified warnings that its plan would fail, during the period when corrective action was still possible. The signals were not invisible. They were visible and ignored.

The mechanics matter. The scope expansion happened in 1992: the system BAE had been hired to build for United at Concourse B was expanded to all three concourses and twenty airlines. BAE's senior managers internally estimated four years; the contract still said two. More than two thousand design changes were issued during construction. The Project Management Team, by its own later admission, halted system design in May 1992 once it "anticipated responsibility for the integrated system" — a clinical phrase that, translated, means the team realized it was going to own a problem it could not solve.

What Denver proves is the premise of every subsequent failure. Warning signs are noticed by engineers, by commissioning staff, by the program manager's deputies, by outside consultants the airport itself hired. They are written down. They reach senior management, or at least the layer below it. And they are filtered, softened, or politically inadmissible by the time they reach anyone who could stop the program.

The mechanism is not corruption. It is something more banal and more durable: the systematic suppression of bad news by an organizational structure that rewards optimism at every reporting layer. The program manager's incentive is to keep the program green. The design-build contractor's incentive is to close out scope. The bond underwriter's incentive is to get to financial close. Each filter is rational. The cumulative effect is that by the time bad news reaches the supervisory board or the airport CEO, four people have softened it to protect client relationships and career trajectories.

This is the architecture of silence. It is more reliable than any of the four signals the failure literature itemizes.

Berlin's Draughtsman

In January 2013, Rainer Schwarz was fired from the chief executive's office at Flughafen Berlin Brandenburg GmbH. He had been reporting, to the supervisory board and to the public, that Berlin Brandenburg Airport was 90% complete and ready to open. Independent auditors had just told the supervisory board that operational readiness sat closer to 56%. The 34-point gap between what Schwarz was reporting and what the building actually was is the architecture of silence at scene level. Schwarz had not invented his number out of nothing. It was the number the program needed in order to keep producing opening dates the political layer above him expected to be told.

BER is the complete specimen. The Bundestag, the federal courts, the city of Berlin, and multiple investigative journalists examined the program in real time, so the institutional failure is documented in unusual depth.

The original budget was €2.83 billion. The terminal opened on October 31, 2020 — nine years late — at a direct cost exceeding €7 billion. When interest and lost revenue are included, the program reaches approximately €10.3 billion. The proximate cause of the failure was the fire protection and smoke extraction system, which failed its mandatory acceptance test in May 2012, twenty-six days before the planned opening.

Here is what was true at that moment. The smoke extraction system had been designed to pump hot smoke downward from ceiling-level ducts through a basement shaft, because the architect had specified a clean roofline without visible exhaust infrastructure. This required reversing the natural buoyancy of heated air at a scale never demonstrated. It is physically possible — but it requires exceptional engineering rigor, and at the dimensions of the BER terminal the system did not work. When the fans built pressure during the May 2012 test, the ducts deformed and the automatic flaps failed to open or close on command. The system's designer, Alfredo di Mauro, was qualified as an engineering draughtsman, not a licensed engineer. His credential status was discovered in 2014. Di Mauro himself later told a Berlin radio station: "No one asked about my university qualifications; that wasn't necessary for the work we carried out."

Read that sentence again. The fire safety system for one of Europe's largest infrastructure projects, the system on which operational certification of the entire terminal depended, was designed by a draughtsman whose credentials no one had verified. That is not a technical failure. That is a governance failure with a physical address.

The supervisory board of Flughafen Berlin Brandenburg GmbH included the mayors of Berlin and Brandenburg, the German Federal Finance Minister, and a mix of civil servants and union officials — politicians, not engineers. Professor Genia Kostka of the Free University of Berlin put it plainly: "The supervisory board was full of politicians who had no idea how to supervise the project. They were in charge of key decisions." When mid-construction scope changes were proposed — north and south piers converting the terminal from a rectangle to a U-shape, an additional retail mezzanine — the body that approved them was making decisions about thermodynamics. Each scope

change cascaded into the fire safety system design, which then had to be re-engineered around a building it had not been designed for.

The program cycled through a carousel of CEOs between 2012 and 2016. Manfred Körtgen was dismissed in May 2012. Schwarz, his successor, was fired the following January for the reporting gap. Hartmut Mehdorn, a turnaround appointment in March 2013, resigned in December 2014 citing political interference. Each new CEO inherited a governance vacuum no single executive had the tenure to fill.

The wiring inspection by TÜV in May 2018 — six years after the failed acceptance test — documented 863 separate wiring defects in the fire and smoke control systems alone. By that point, 66,500 total defects had been catalogued across the terminal, of which 5,845 were classified as critical. The fire safety system did not receive TÜV approval until April 2019. The first full-systems test rehearsal began August 2019. The airport opened fourteen months later, into a pandemic that masked the most consequential downstream failure — the business case itself.

The second collapse at BER was not the building. It was the commercial assumption underneath it.

The terminal had been designed for 34 million passengers annually at opening, with capacity for 46 million. The retail revenue model assumed half of total airport income from concessions, calibrated to the spending behavior of premium connecting passengers — the volume Air Berlin's planned Oneworld connecting hub was supposed to generate. Air Berlin filed for insolvency on August 15, 2017, three years before BER opened. The carrier's financial distress had been visible in its public filings since at least 2015. Airport leadership publicly stated in 2017 that the bankruptcy "would have had no imminent impact on expected traffic volume." That sentence is technically defensible and commercially absurd. The connecting hub model that underwrote half of BER's retail revenue dissolved with the carrier that was supposed to feed it.

BER's first full year of operations, in 2021, saw 9.95 million passengers against a design calibrated to 34 million. By 2023 the airport had recovered to 23.1 million — still below the 34 million the commercial design required. The carrier mix had shifted decisively: Ryanair and easyJet together control roughly a third of total seat capacity. BER is now what its actual market always was — a low-cost, origin-and-destination airport — and not what its business case said it would be.

The signals at BER were visible. The draughtsman's credentials. The smoke extraction design that required reversing natural buoyancy at unprecedented scale. The supervisory board's composition. Air Berlin's deteriorating balance sheet. Schwarz's 90% report against the auditors' 56%. Every one was knowable and known. What failed was not detection.

The North American Echo

Mirabel is the cleanest historical case in the brief base, and it sits unread in most modern terminal post-mortems because it is so old. Montreal–Mirabel opened in 1975 as

Canada's eastern gateway, sized for 20 million passengers a year. It never carried more than 3 million. The federal government's traffic forecasts had been built on technological assumptions about transatlantic refueling stops that longer-range jets made obsolete almost immediately after groundbreaking. The terminal was demolished in 2014.

Mirabel is what Flyvbjerg calls optimism bias in its pure form — a forecast that the world had already invalidated by the time the asphalt was poured.

BER's commercial collapse has a US precedent, closer to the current capital cycle than the German specifics suggest. The earlier US wave — 1989 to 1998, the largest sustained airport capital program until the present moment — produced Denver, Pittsburgh, and Atlanta's Hartsfield international terminal. Its defining characteristic was optimism about carrier stability. Its defining failure was designing to a single-carrier hub business case that airline bankruptcy could invalidate.

Pittsburgh is the case to sit with. Its 1992 terminal was purpose-built for US Airways to handle 32 million passengers annually. By the late 1990s the airport peaked at 20 million — 80% of them connecting. When US Airways entered bankruptcy twice between 2002 and 2004 and dehubbed Pittsburgh in December 2004, traffic fell from 20 million to 8 million. The airport was left managing facilities built for four times its actual traffic. County bonds worth \$600 million, expected to be serviced by the airline's 30-year lease, had to be restructured using state gaming revenue and Marcellus shale mineral leases. The terminal — once called "airport of the future" by the New York Times — became the most visible symbol of what happens when a building is sized to one carrier's specifications and that carrier's network strategy changes.

Memphis ran the same play with a faster collapse. Northwest Airlines built Memphis into a connecting hub from the mid-1980s. Delta acquired Northwest in 2008. By 2013, Delta had concluded that Memphis was redundant to Atlanta — same Southeast origin base, lower-yield connecting traffic. Delta announced the dehub in June 2013 and executed by September. Passenger traffic fell from 11.3 million at the 2007 peak to 3.5 million by 2015. Cincinnati, St. Louis, and Cleveland sit in the same pattern band — five US airports built or expanded to a hub carrier's specification and dehubbed within the decade.

This is the BER failure mode in US form. The terminal works. The building is sound. The business case dissolves because the airline strategy that justified it changed. Any current US terminal program with significant connecting-traffic exposure to a single carrier — JFK NTO's reliance on Air France, Lufthansa, Turkish, and Korean; O'Hare's continued bet on a stable United–American duopoly — is exposed to the same failure mode regardless of how well the building is delivered. The asymmetry is the load-bearing point: the airport builds; the airline decides whether to fly. Atlanta's international terminal is the success case of the earlier wave not because Delta's commitment was guaranteed, but because the airport's use-and-lease was structured so that Delta paid into the building as Delta grew into it. The carrier signed up to the geometry. That is the difference.

Six Months Was What Was Agreed

"I remember suggesting nine months, but six months was what was agreed, and built into the programme." The line is Ian Booker's, the specialist systems engineer responsible for testing at Heathrow Terminal 5. Building magazine reported it in the post-mortem. Booker's recommendation was the entire argument compressed to a sentence — and the override that followed was the architecture of silence in working order.

T5 cost £4.2 billion and was, by the standards of British construction practice, a triumph. The building was delivered on time and on budget under BAA's bespoke "T5 Agreement," a relational contract in which BAA retained all risk on its own balance sheet and paid contractors actual cost plus a fixed fee. There were no incentives for change orders or claims. The supply chain was integrated under genuine in-house program management. The OECD's Olympic Infrastructure Toolkit cites T5 as the canonical example of how to make construction work.

It opened on March 27, 2008. Within hours, British Airways suspended hold baggage check-in and cancelled 34 flights. Over the next ten days, more than 42,000 bags failed to travel with their owners and over 500 flights were cancelled. The root cause has been documented in detail by the UK Parliament's Transport Select Committee. It was not the building. It was a software filter, left in place from the testing phase, that prevented the live baggage system from recognizing interline bags transferring from other airlines. The filter had been inserted specifically to prevent test messages from contaminating production systems and was never removed at go-live. Bags arriving from other carriers were automatically routed to manual storage. Staff numbers were not sufficient to clear the backlog. The system clogged.

This is the kind of failure that only surfaces in integrated, end-to-end testing under live conditions. And the testing window had been compressed. Booker had recommended nine months of operational readiness testing. The program delivered six. Building magazine documented that testing ran "half as long as recommended." Construction delays consumed the integration window. By the time British Airways could begin on-site testing — October 31, 2007, five months before opening — the proving trials had been reduced in scope or cancelled because the terminal was still an active construction site. BA's on-site familiarization program for ramp and passenger service staff was deferred by six weeks.

The opening morning is documented in granular detail. BAA had not made the staff car park accessible; British Airways baggage handlers arrived two hours late. The staff access system did not recognize employee IDs. Eighteen terminal lifts were jammed. The Heathrow Express and the transit connecting the main terminal to the satellite building were down. These are not separate failures. They are a single failure — the operational systems that ran the building had never been tested in concert with the staff who ran them, because the integration window had been compressed below the duration the testing engineer said the work required. Booker said nine months; the program said six. The decision was on paper, in the program schedule, from the moment it was made.

T5 recovered, and the recovery was real. By February 2009 — within eleven months of opening — BA's daily baggage mishandling rate at T5 had fallen to five bags in a thousand. By 2018, the terminal was processing 32.1 million passengers, above its 30-million design capacity. From 2012 through 2016 it won the Skytrax Best Airport Terminal award for five consecutive years. The terminal that began as a national humiliation became, in operating performance, one of Heathrow's strongest assets.

But recovery is not absolution. BA's director of operations and director of customer services were both terminated. Citibank estimated direct cost to BA at \$50 million. The brand damage sustained for a period; BA's Heathrow operation became, briefly, an international punchline. The failure was preventable in a specific, named way. Someone with standing — Booker had it on the testing program; nobody had it on the integration calendar that overrode him — could have refused the compression. None did.

T5 produced a falsifiable hypothesis the current US wave is now testing. JFK New Terminal One's Phase A is scheduled to open no earlier than November 2026. SITA, the Master Systems Integrator, opened a pre-production testing center near Islip, Long Island, in December 2023 — three years before the target opening. JLL was engaged in 2025 for operational readiness work. The structure is right; the schedule is tight. If NTO Phase A opens in November 2026 with baggage mishandling below T5's day-one rate — roughly 4,200 bags per day in the first ten days — and CBP processing within design tolerance, the institutional check held. If the system fails at integration the way T5's did, the integration window was again compressed below what the engineers responsible for testing requested. We will know by year-end 2026.

The Concession Trap

In August 2019, Denver International Airport terminated its 34-year design-build-finance-operate-maintain contract with Great Hall Partners. The project had been awarded eighteen months earlier, in 2018, at approximately \$650 million. By month eleven, GHP — led by Ferrovial Airports — had issued change order notifications totaling \$288 million. Structural concrete defects had been discovered mid-construction. A city council member who had authorized the original 2017 vote later called it "the worst vote he'd ever made." The independent review found that DIA had managed the DBFOM contract as if it were design-bid-build — issuing owner changes, inserting itself into decisions the contract had explicitly delegated to the private partner, while simultaneously accusing the contractor of scope creep.

The project was repackaged and approved at approximately \$2.35 billion, with completion now scheduled for 2028 — seven years beyond the original schedule. The post-mortem concluded that the project "may have been better served as a design-bid-build contract or a construction manager/general contractor project." The institutional learning was paid for in eight years of delay and roughly \$1.7 billion of cost growth.

The finding indicts the procurement structure rather than the contractor or the airport executive. The institution chose a contract form that did not match its own operating capacity. The choice was made for political and financing reasons — the city wanted the

project off its balance sheet, and a 34-year concession provided that — and the operating consequences were absorbed by the airport that still had to manage the program day to day.

This is the new failure mode of the P3 era, and it is downstream of something larger. The PFC has been frozen at \$4.50 since 2000 — worth \$2.72 in 2018 dollars, a 40% real reduction across six FAA reauthorization cycles. ACI-NA estimates \$151 billion in airport project financing needs over 2023–2028 against combined annual federal and PFC revenue of roughly \$12 billion, leaving approximately \$18 billion per year of unfunded need. To fill that gap, airports use revenue bonds, special-facilities revenue bonds, private activity bonds, and P3 concessions. Each structure introduces governance complexity. The PFC cap forced LaGuardia Terminal B and JFK New Terminal One into special-facilities and demand-risk P3 structures. The financing structures then forced the contract structures that diffused decision authority.

There is a regulatory floor underneath this that compression cannot cross. The FAA's 14 CFR Part 139 governs certificated airports. Every change to construction sequencing that affects runways, taxiways, aprons, or ARFF response routes must run through the airport's Construction Safety and Phasing Plan and be coordinated with the FAA regional office. Non-compliance is a certificate-jeopardy event. A schedule "optimization" made in the project office does not automatically become executable on the airfield. The owner who does not understand this will be told by the regional office, and the schedule will reset.

The LAX Automated People Mover is the live counterexample to the proposition that P3 structures discipline costs. Awarded as a 30-year design-build-finance-operate-maintain contract, the project was originally budgeted at \$1.9 billion. The current estimated total project cost is \$2.78–3.34 billion. The 2024–25 Los Angeles County Grand Jury found that the contractor, LINXS, "leveraged the change order process by implicitly holding out the threat of prolonged litigation to force LAWA to agree." Los Angeles World Airports paid more than \$200 million in dispute settlements through 2023, then agreed to a \$550 million global settlement in 2024. Fitch downgraded the project's credit, citing "extended construction delays, prolonged dispute resolution, and difficulties in the parties' working relationship." The P3 structure did not transfer the risk. It transferred the risk of litigation expense back to the public owner through a conflict resolution mechanism the contractor learned to use against it.

JFK New Terminal One is the live test. The \$9.5 billion program is the largest single-asset P3 in US history. SITA serves as Master Systems Integrator within an alliance that includes AECOM Tishman, Schneider Electric, and Faith Group — a structure designed to address the integration risk that has historically broken comparable programs. The pre-production testing center near Islip, opened in December 2023, is the right architecture.

The warning signs are also present. The Phase A opening, originally targeted for summer 2026 and timed to the FIFA World Cup, was pushed to no earlier than November 2026.

Moody's revised the project's senior secured outlook to negative in May 2026, citing the postponement and a downward revision in forecasted passenger demand. The January 2025 Alternative Solutions Report described the project as operating in a "recovery posture rather than comfortable buffering." The credit structure embeds material demand risk: 23 million passengers at full build-out is required to validate the cost structure, with no Port Authority general revenue backstop. If the international carriers that make up the tenant base reduce JFK frequency for any reason, NTO's debt service coverage compresses with no public buffer.

The World Cup is the political opening date. FIFA matches in June and July 2026 create exactly the fixed external deadline BER's supervisory board kept committing to and the actual systems work could not meet. NTO is now testing whether the institutional check — the Port Authority, the Ferrovial-led consortium, SITA, JLL — has the standing to refuse an opening date the systems are not ready for. The November slip suggests it has, so far. The remaining question is whether the November date holds.

The 482-Foot Walk

A passenger making an international connection at the new LaGuardia Terminal B walks four hundred and eighty-two feet across an elevated pedestrian bridge sixty feet above the apron, descends to the satellite concourse, makes the connection, and walks back. The transit, end to end, takes about twenty-two minutes on a good day. The previous Terminal B's equivalent transfer took roughly seventeen. The geometry is permanent. It is also award-winning: Terminal B opened in 2022, won the 2023 Skytrax World's Best New Airport Terminal award and the ACI Service Quality Award, and is by most measures a success.

The airside gain is real. Airport-wide flight delays fell from approximately 26.6% to 20.75% — a 22% reduction in delay rate that traces directly to the airside geometry change. The new terminal recovered 40 acres of airside land and added approximately two miles of taxiway, giving aircraft two independent paths to each of 35 gates.

And yet. The satellite concourse design extended walking distances. International connection times rose by approximately 25%. The drop-off zone, with a single road access, regularly produces wait times above 20 minutes. The proposed AirTrain landside connection — the offsetting move in the original business case — was cancelled in March 2023 after estimates rose from \$450 million to \$2.4 billion, a 433% overrun before a single dollar of construction. The Port Authority substituted an enhanced bus strategy. The landside assumption embedded in the original Terminal B business case did not survive contact with cost estimating.

LaGuardia Terminal B is the case where the building works, the airside works, and the operational performance the building enables is worse on multiple dimensions than what it replaced. The signals the standard framework reads — scope creep, ORAT compression, owner-engineer confusion, optimism bias — do not all appear. What appears instead is a building whose connecting-passenger experience and curbside

experience worsened, and whose offsetting investment was cancelled at a 433% overrun before construction. By the time ORAT runs, the geometry is fixed.

The connecting passenger making a 22-minute international transfer at the new LGA is paying for decisions made several years earlier by people who would never have to walk a connecting passenger through the building on a Monday morning. The way an operator catches this failure is not by watching the schedule. It is by having an operating organization with standing to require that someone who will run the building sign off on the circulation design before the architect freezes it. Most terminal programs do not have that organization. They have program management firms whose incentive is to close out scope.

Two Years of Trials

Istanbul Airport opened in October 2018. The ORAT program ran more than two years. Advisers from Copenhagen Airport and Incheon Airport, both of which had managed comparable greenfield openings, were brought in to test scenarios the local team had not encountered. Multiple full-scale live trials were conducted with real Turkish Airlines aircraft and thousands of test passengers and baggage items. The 42-kilometer baggage handling system was rated at 28,880 bags per hour. During the first summer peak in 2019, system availability reached 99.9% and average in-system bag time was approximately 8 minutes against a 15-minute design specification — a 47% outperformance on spec.

Singapore Changi Terminal 4 is the case to sit inside for a moment, because the testing program is the entire point. T4 opened in October 2017. Before it did, Changi Airport Group conducted more than 100 trials. 2,500 staff and 7,500 volunteers participated. The volunteers were the public — Singaporean residents who arrived at the new terminal in waves, checked in with real airlines using real boarding passes, dropped real bags into the new BEUMER CrisBag sortation system, queued at biometric immigration gates, boarded test aircraft. They did this across months. The biometric architecture — Fast and Seamless Travel, deployed across check-in, bag drop, immigration, and boarding — was stressed under conditions resembling the airport it would actually become. The scope was fixed before testing began. The system was designed around a defined passenger experience specification, not retrofitted to a building that had changed mid-construction. The ORAT process was led by the operator who would be accountable for post-opening performance — not by a construction contractor measuring completion against milestones. There is no documented baggage system failure or significant operational incident in T4's first year.

These two cases share a structural feature none of the failure cases share. Scope was locked before commissioning began. ORAT was funded as a multi-year program with its own budget line. A single integrating authority — Changi Airport Group, İGA — held responsibility for both construction completion and operational performance. The airline customers had signed off on what they were getting before design was frozen.

The distinction is not that the successful cases avoided the four warning signals. Every megaproject of this complexity sees scope pressure, schedule pressure, and optimism in the early business case. The distinction is that the successful institutions built governance structures that could refuse compromise. Someone with standing could say, two years before opening, that ORAT would run on a fixed multi-year calendar and that no political opening date would compress it. Someone could insist airline tenants sign legally binding commitments before the bond was issued, rather than letters of intent that protected no one.

These are not warning sign detectors. They are warning sign actors. The difference between the failure cases and the success cases is not who could see the signal. It is who could act on it.

The Failure Clock That Keeps Running

The Year 1–3 failures get the cinema. The Year 10–15 failures get the deferred maintenance backlog.

Denver's automated baggage system ran for ten years before decommissioning in 2005, at roughly a million dollars a month in maintenance — a Year 10 cost driven by the same optimism bias that produced the original specification. The conventional manual system that replaced it could have been built for \$51 million at the start. United Airlines absorbed the difference across a decade.

Terminal capital expenditure represents only 15–20% of a 30-year lifecycle cost. Baggage handling system CAPEX accounts for 30–40% of its lifecycle cost. Operating expenditure — energy, maintenance contracts, spare parts, labor — consumes 60–70% of the total. A sponsor who optimizes the ribbon-cutting price and defers decisions about maintenance funding, energy performance, and equipment service contracts is pre-loading failure at the 10–15 year mark, when major mechanical systems hit their first replacement cycle.

Lifecycle cost thinking is structurally absent from most mega-terminal business cases. There is no public evidence that the current US wave has corrected for it. Optimism bias at Year Zero produces an underfunded capital reserve at Year Ten. The cost-per-enplanement ratchet that no individual operator can reverse is the long-tail consequence. That is its own architecture of silence — quieter than BER's because it produces no opening-day headline, materially more expensive in present-value terms over a 30-year asset life.

The Counter-Case, Honestly Presented

A serious reader should not leave without seeing this argument's weakest joints. There are four.

The first is that the warning signs are not diagnostic. Flyvbjerg's database of over 16,000 projects across five decades finds that only 8.5% of megaprojects meet time and budget targets, and only 0.5% deliver on time, on budget, and with promised benefits fully

realized. If scope creep, optimism bias, and systems integration deferral appear in 91.5% of megaprojects, they cannot distinguish failures from marginal successes. A variable that predicts everything predicts nothing. A warning sign is useful only if its absence correlates with success. The thesis as initially stated did not prove that absence.

The second is harder. The essay studies a handful of publicly known failure cases — BER, Denver, T5, LGA — alongside a handful of publicly known success cases — Istanbul, Changi T4, DFW Terminal F. The success cases were chosen partly because they succeeded; the failure cases became visible partly because they failed. Selection bias at this scale is not solvable from the brief base. It is a real limit on what the argument can claim.

The third is that the four cases are not the same failure. BER took nine years to recover and required structural rebuilding of a building that did not work. T5 recovered in eleven months and required no structural change. LaGuardia Terminal B opened on time and won an architectural prize while delivering a 22-minute international connection where the old terminal delivered seventeen. JFK New Terminal One has slipped six months on a system not yet open. Calling these the same pattern because they all involve "systems integration deferral" or "scope creep" is like saying four patients died of "circulatory failure" — technically accurate, clinically useless. A framework that treats a six-month slip and a nine-year disaster as members of the same set will mislead the executives it is designed to help.

The fourth is the most uncomfortable. Optimism bias is not a correctable governance failure. It is a rational response to a funding environment that rewards ambition and penalizes candor. Flyvbjerg's research on strategic misrepresentation documents that promoters who present realistic forecasts often do not get funded. The political economy that produces billion-dollar terminal programs also produces the optimism that makes them approvable. The PFC cap has been frozen at \$4.50 since 2000 — six FAA reauthorization cycles with the same result. CBP staffing is appropriations-driven; major international airports document processing times two to three times normal when officers are redeployed. TSA changes checkpoint design standards mid-construction. The DCA perimeter rule has survived sixty years of economic arguments against it. The immovable federal constraints are what produce the financing structures — P3s, special-facilities revenue bonds, demand-risk debt — that produce the governance fragmentation. Telling airport executives to be less optimistic is like telling structural engineers to use less gravity.

The argument as initially advanced overreaches. The honest question is whether the corrections leave the core thesis standing.

Why the Counter-Case Is Insufficient

They do.

Take the universality objection at full strength. The signals are universal at the design phase. What is not universal is whether the institution receiving the signal has the standing to act on it. That is the actual variable. The success cases did not have categorically different signals. They had categorically different institutional architecture for receiving and acting on them.

The decision rule has to be sharper than "be vigilant," precisely because the signals are nearly universal. The rule the success cases imply: trigger action when any one of the four signals appears *and* no compensating institutional check exists. Scope changed mid-construction? Acceptable only if the use-and-lease agreement is reopened, the bond covenant is re-disclosed, and the integrated testing schedule is re-baselined. ORAT compressed below the four-month IATA floor? Acceptable only if a board-level covenant prohibits an opening date within four months of integrated testing completion. The framework does not say "stop the project" when a signal appears. It says "force a pause and a re-baseline, or accept that the institutional check is missing and write down what that costs."

The selection-bias objection is harder, and the argument is partly vulnerable to it. The institutional-standing reformulation cannot be proven on the brief base alone; it can only be made plausible. What raises its plausibility above coincidence is that the structural features the success cases share — fixed scope before commissioning, an integrating authority spanning construction and operations, multi-year ORAT funded as a budget line rather than a schedule activity, airline tenants legally bound before bond issuance — were written into the contract structures and use agreements at the start. That is a falsifiable claim about the design phase, not a story told in hindsight. If a future failure case is found that had all four of these structural features in place from the start, the argument is in trouble. The argument can be wrong. It is not currently known to be wrong.

The categorical-confusion objection forces an adjustment. BER, T5, and LaGuardia Terminal B are different categories of failure that share a common pre-condition. BER is structural: the building did not work and could not be made to work without years of remediation. T5 is operational: the building worked but the human and information systems running it failed at integration. LaGuardia Terminal B is a third category: the building works and parts of the operation it enables are worse than what it replaced. The four-signal framework is a screening test for the underlying institutional condition, not a predictor of which failure category will manifest. A program that compresses ORAT may produce T5. A program that lets scope expand without re-baselining may produce BER. A program that excludes operating staff from design review may produce LaGuardia. The category is downstream of the institutional architecture.

The optimism-bias objection forces the hardest concession. Optimism bias is structural. Airport executives cannot raise the PFC, cannot control CBP staffing, cannot freeze TSA checkpoint standards, and cannot eliminate the political pressure for opening dates. They can choose their own governance architecture. They can choose whether the use-and-lease is renegotiated before the bond is issued. They can choose whether ORAT is a

budget line with a two-year horizon or a schedule activity that gets compressed when construction slips. They can choose whether integrated systems testing is built into a contractual covenant the program manager cannot waive under schedule pressure. Reference-class forecasting — Flyvbjerg's method, which applies the base rate of comparable projects to your project's estimate — was adopted by the UK Treasury more than fifteen years ago [UNVERIFIED — HUMAN REVIEW]. Almost no US airport uses it. A board that knows its business case is optimistic and does nothing structural to compensate is choosing to walk into the predictable failure.

The original thesis was right in its observation and incomplete in its prescription. The signals are visible. They are also universal. What predicts the actual outcome is the architecture of who can act on them. Munich's engineers warned Denver. Booker said nine months. The TÜV inspectors flagged the BER fire system year after year. Schwarz reported 90% completion against the auditors' 56%. The signals arrived. They were not heard by anyone with the authority and the will to halt the program.

The success cases prove the inverse. At DFW, Sean Donohue closed a 10-year use-and-lease agreement before committing the airport to a \$9 billion capital program. Terminal F expanded from \$1.6 billion to \$4 billion only because American Airlines voluntarily increased its commitment. The scope growth was carrier-driven. Donohue did not stop scope creep; he made scope creep someone else's risk. That is what an institutional check looks like when it is built before construction begins.

The signals are not the framework. The framework is the architecture you build to act on them.

Implications for MWAA and the Dulles Transformation

This essay was commissioned by the operator of one of the largest forthcoming capital programs in US aviation. The Metropolitan Washington Airports Authority is the right case to apply the argument to, and the choice is not arbitrary. Three reasons.

First, the Dulles transformation under evaluation is in the same scale band as the failure cases this essay examines — current cost estimates approach \$22 billion, materially larger than the \$13.7 billion combined cost of JFK NTO (\$9.5 billion) and JFK Terminal 6 (\$4.2 billion), and roughly twice O'Hare's \$11.5 billion expansion. Second, MWAA's federally chartered governance structure sits between the political-board pathology of BER and the independent-authority strength of Port Authorities — meaning the institutional architecture question is live, not settled. Third, MWAA has already produced one direct precedent that the failure literature catches.

The Dulles Aerotrains is the precedent worth re-reading. The \$1.4 billion system opened in January 2010, four months behind its originally planned fall 2009 opening, linking Concourses A, B, and C — but not Concourse D. The four-month delay reads two ways. One read is operational discipline: MWAA held the opening to complete reliability testing rather than open a system the integrated trials had not finished. The competing

read is scope failure: the 1960s-era mobile lounges the Aerotrains were meant to replace still serve Concourse D today, fifteen years on. MWAA paid \$1.4 billion for partial coverage of a system that still requires the supplemental infrastructure it was meant to retire. The \$22 billion transformation now in evaluation includes the rebuild of the mobile lounge dependency. The lesson is not that the Aerotrains were a failure — it was not — but that scope decisions made for political and financing reasons will outlast the people who made them and become operating costs the next generation absorbs.

Four directives follow.

One: name the integrating authority before selecting the contracting structure. Before any procurement form is chosen — design-bid-build, CMAR, P3 — name the person who will hold integrated systems responsibility from basis-of-design through Year 3 of operations. Not a program manager. Not a developer. A senior MWAA executive with operational accountability and board-level standing. The Denver Great Hall failed because the airport managed a DBFOM as design-bid-build. The institutional architecture must precede the contract form, not the other way around. The arithmetic is concrete: at MWAA's current rating posture, a 30–60 basis point spread widening on a \$1.5 billion bond issuance is \$4.5–9 million annually over a 30-year tail. The cost of an integrating authority sits below that number by an order of magnitude. The cost of not having one sits above it. No contract form by itself produces the authority — but every success case names the person, with the scope, before the contract is signed.

Two: sequence the airline use-and-lease before committing to scope. Donohue did this at DFW. MWAA's existing use-and-lease structure is the framework in which a \$22 billion program will be priced into cost per enplanement, and the network carriers — United principally at Dulles, but the carriers routing through IAD as a function of slot strategy at DCA — will either absorb the cost increase or reroute. The math is concrete: a \$500 million bond program adds approximately \$0.80 to CPE at a 40-million-enplanement airport, and 12% CPE increases historically trigger formal rate challenges under most use agreements. United at Newark has already begun shifting capacity to Dulles in response to EWR's \$26.17 CPE. That is opportunity for IAD only if IAD's post-program CPE does not produce the same response in reverse. The use-and-lease cannot be negotiated after the bond is issued and the scope is fixed. It must precede them.

Three: build the ORAT program — and a board-level covenant protecting it — into the construction baseline. IATA's documented minimum is four months of integrated operational trials before opening. Istanbul ran ORAT for more than two years. Changi T4 ran more than 100 trials with 2,500 staff and 7,500 volunteers across many months. T5 compressed to six months, against the nine its lead testing engineer requested. BER compressed integration testing repeatedly across nine years of postponements. The decision to fund ORAT at the success-case duration must be made when the construction schedule is approved, not after construction slippage has consumed the integration window. The institutional check the failure cases lacked is a person insulated from the political opening-date commitment, whose performance is

measured by surfacing what will break before opening rather than by hitting milestone celebrations. A board that does not designate this function in advance is choosing, by omission, to defend the opening date the day the system fails its acceptance test.

Four: take the federal constraints as fixed and design around them. The PFC will not move in this reauthorization cycle. The IIJA \$19.5 billion supplemental aviation pool expires in December 2026; baseline AIP reverts to roughly \$4 billion annually. CBP staffing is appropriations-driven; the international terminal business case must be modeled against the CBP staffing that is funded, not the staffing the design assumes. The TSA checkpoint footprint must be sized for the CT scanner standard, not for the 2D X-ray standard current when feasibility study began. JFK NTO designed its arrivals hall around CBP's Enhanced Passenger Processing biometric program precisely because traditional FIS staffing could not be assumed at design capacity. That is the right pattern. Reference-class forecasting — applying the base rate of comparable projects, not the program-specific feasibility numbers, to the financing assumption — is the discipline most US airports do not yet practice. If the MWAA financing structure cannot survive an 80% cost overrun (Flyvbjerg's base rate for megaprojects) and a major Year 1 demand shortfall, then the financing structure is not the right one.

These are not exotic recommendations. They are what the success cases exhibit and what the failure cases lack. They are also more difficult to implement than they sound, because each runs against an institutional incentive that exists for rational reasons. The airline use-and-lease takes years to renegotiate. The board-level ORAT covenant looks like a constraint on management flexibility. The designated integrating authority is uncomfortable to work with. Reference-class forecasting produces estimates political authorities do not want to publish.

The question MWAA — or any authority sitting before a \$20 billion capital decision — has to answer is the one Munich's engineers asked Denver in 1992 and Denver did not answer. If your own commissioning engineers told you, in writing, that the program as scoped would fail, what is the institutional pathway by which that signal reaches the person who can stop it? If the pathway requires more than three handoffs, the signal will not arrive in time. The architecture of silence will absorb it.

There is a photograph from BER's October 2020 opening day. The terminal is largely empty. A handful of passengers move through a vast hall designed for thirty-four million annual travelers. The pandemic is responsible for most of the emptiness, and also, perversely, for the fact that the building's first weeks did not produce a Year 1 catastrophe. The systems that had failed for nine years did not have to perform at design load. The retail concessions calibrated for premium connecting passengers were closed. The terminal had opened. There was no one there to see it. Somewhere in Berlin, the planners who had committed to a 2011 opening, then 2012, then 2013, then 2014, then 2017 — the people who had received the warnings and let them pass — would have looked at that photograph and recognized something it took them nine years to admit:

the building they had finally finished was no longer the building the business case had been written for.

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